Bipartite graphs of diameter 7 from the split Cayley hexagon: level graphs, surgery and hubs

Degree–diameter lab (Claude Code sessions coordinated by Doğa Özden)

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Draft of 25 October 2026, written for Ignacio López and Gabriela Araujo-Pardo by the degree–diameter lab (github.com/dogaozden/degree-diameter; Claude Code sessions coordinated by Doğa Özden).

This note states and proves the constructions behind the lab's new lower bounds for the bipartite degree–diameter problem at diameter 7 (cells (7,7), (9,7)–(16,7)). Everything below is meant to be checkable by hand: the geometric hypotheses are stated in the language of the hexagon, and the neighbourhood of every vertex of every graph is given explicitly (Definitions 2.1 and 3.1). Where a step depends on a computer search (the specific large sets and the auxiliary "hub graphs" behind the record values), we say so and give the check. Section 10 lists the computer tests of every theorem on small q.

File paths refer to the repository github.com/dogaozden/degree-diameter. Unless marked otherwise they are on the master branch (commit c48c976); [b14] marks files on branch claude/cell-14-7-record-5d03gb and [b15] files on branch claude/cell-15-7-haar-search-x6ynhl. This note and its tests (research/theorems/D7_HEXAGON.md, d7hex_tests/, d7hex_lemma41/) are on branch claude/d7-hexagon-theorems-4w1prt. Section 12 collects remarks for readers of the lab's per-cell notes.

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0. Results

Notation: $n_b(\Delta, D)$ is the largest order of a bipartite graph with maximum degree $\leq \Delta$ and diameter $\leq D$. For a prime power q put

$$N = N(q) = \frac{q^6 - 1}{q - 1} = (q + 1)(q^4 + q^2 + 1),$$

the number of points (and of lines) of a generalized hexagon of order (q, q). In the 2013 table [MS13, Table 4] each cell below has the value $(2r + 1) \cdot 2N(q)$, with $\Delta = q + 1 + r$ for the (q, r) listed (the largest value of this form for that Δ); this is the order of the level graph of Section 2.

cell $(\Delta, 7)$	q, r	table (2013)	ours	construction (this note)	parameters
(7,7)	5, 1	23,436	**23,598**	3 levels, one local duality (Cor. 8.1)	$\|R\| = 81$ (72 pairwise opposite + 9)
(9,7)	7, 1	117,648	**118,238**	same	$\|R\| = 295$ (288 + 7)
(10,7)	8, 1	224,694	**225,152**	same	$\|R\| = 229$ (217 + 12)
(11,7)	9, 1	398,580	**413,180**	3 levels, duality, spread points freed (Cor. 8.2)	$\|F\| = 7{,}300$
(12,7)	9, 2	664,300	**693,500**	5 levels, duality, spread points freed (Cor. 8.2)	$\|F\| = 7{,}300$, $14{,}600 + 14{,}600$ hubs
(13,7)	11, 1	1,062,936	**1,064,472**	3 levels, one local duality (Cor. 8.1)	$\|R\| = 768$ (756 + 12)
(14,7)	11, 2	1,771,560	**1,789,588**	typed levels HDHDH + local duality (Cor. 8.3)	8,645 pts / 8,246 lines / 768; $9{,}014 + 9{,}014$ hubs
(15,7)	11, 3	2,480,184	**2,501,491**	typed levels HHDHDHD + local duality (Cor. 8.3)	twelve segment sets, 768; $10{,}826 + 10{,}481$ hubs
(16,7)	13, 2	4,022,340	**4,052,108**	typed levels HDHDH + pairwise opposite sets (Cor. 8.3)	14,762 pts / 14,823 lines / 122; $14{,}884 + 14{,}884$ hubs

All nine graphs were rebuilt by the lab's coordinator from their solution files and checked by a breadth-first search from every vertex (no symmetry assumed): bipartite, maximum degree Δ, diameter 7 (logs in research/coordinator_check/, Section 10). The cell (8,7) record is a Haar graph and is not discussed here.

Our comparison is with the 2013 table only. We could not read the current CombinatoricsWiki table from our environment (the page returns HTTP 503), and a literature search found no later values for these cells; we would be grateful to hear of any.

The theorems are:

- **Theorem A** (classical): the level graph with $2r+1$ levels and arbitrary matchings has diameter ≤ 7.
- **Lemma 3.3 and Theorem 3.4** (assembly): if every "transition" survives the deletion of its freed matching edges with all its pairs still within distance 7, then hub vertices can be attached to the freed slots in any way, and the result has diameter at most 7 as soon as hubs of the same colour are within distance 6.

- **Theorem C**: a transition whose matching is the restriction of an automorphism of the incidence graph (the identity between H and its dual, or a duality of H) survives the deletion at any *segment set*: a point set in which every point is collinear with other points of the set along at most one line. Such sets have at most $(q+1)(q^3+1)$ points, with equality exactly for the point sets of spreads; and a transition of this kind is repaired only if its freed set is a segment set (Remark 4.2).
- **Theorem B**: a transition with an arbitrary matching M survives the deletion at a non-collinear point set R if $M(R)$ is non-concurrent and M maps the points collinear with x onto the lines concurrent with $M(x)$, for every $x \in R$. Propositions 5.1 and 5.2 say when such an M exists ("mirrored sets").

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1. The split Cayley hexagon

1.1 Coordinates

Let q be a prime power and $V = \mathbb{F}_q^7$ with coordinates $(X_0, \ldots, X_6)$. Let

$$Q(X) = X_0X_4 + X_1X_5 + X_2X_6 - X_3^2,$$

with polar form $b(X,Y) = Q(X+Y) - Q(X) - Q(Y) = X_0Y_4 + X_4Y_0 + X_1Y_5 + X_5Y_1 + X_2Y_6 + X_6Y_2 - 2X_3Y_3$ (alternating when q is even). The parabolic quadric $Q(6,q)$ is the set of points $\langle x\rangle$ of $\mathrm{PG}(6,q)$ with $Q(x) = 0$. It is convenient to write $x = (v, a, w)$ with $v = (x_0, x_1, x_2)$, $a = x_3$, $w = (x_4, x_5, x_6)$; then $Q(x) = v \cdot w - a^2$.

The **split Cayley hexagon** $H = H(q)$ ([Ti59]; see [VM98, §2.4]) has

- **points**: the set P of all $N = (q^6-1)/(q-1)$ points of $Q(6,q)$;
- **lines**: the set L of the lines of $Q(6,q)$ whose Grassmann coordinates $p_{ij} = x_iy_j - x_jy_i$ (for any two spanning points x, y) satisfy Tits' equations, which in the labelling of coordinates used here read

$$p_{12} = p_{34}, \quad p_{54} = p_{32}, \quad p_{20} = p_{35}, \quad p_{65} = p_{30}, \quad p_{01} = p_{36}, \quad p_{46} = p_{31};$$

- **incidence**: containment.

(Sources label the coordinates differently, and the equations change accordingly; this is the form used by all the lab's generators. We checked by computer, from scratch, that with this quadric it gives a generalized hexagon of order (q,q) for $q = 2, 3, 4, 5$: right numbers of points and lines, $q+1$ on each, incidence graph of diameter 6 and girth 12, Section 10.1.)

Example. For $x = e_0 = (1, 0, \ldots, 0)$ and a point y, $p_{0j} = y_j$ and $p_{ij} = 0$ for $i, j \geq 1$; the equations reduce to $y_1 = y_2 = y_3 = 0$, and $b(e_0, y) = y_4 = 0$. So the $q+1$ lines through e_0 are $\langle e_0, \lambda e_5 + \mu e_6\rangle$, $(\lambda : \mu) \in \mathrm{PG}(1,q)$: they lie in the totally singular plane $\langle e_0, e_5, e_6\rangle$. (In general the lines of $H(q)$ through a point x are the lines through x of a totally singular plane $\pi_x \subset x^\perp$.)

$H(q)$ is a generalized hexagon of order (q,q): N points and N lines, $q+1$ points on a line and $q+1$ lines on a point. The group $G_2(q)$ acts on it; for instance $g \in SL(3,q)$ acts by $(v,a,w) \mapsto (gv, a, g^{-T}w)$ (this is the symmetry used to check the (16,7) graph, research/squads/cells/d16D7.md).

The **dual hexagon** H^D has the lines of H as points and the points of H as lines. A **duality** of H is an isomorphism $H \to H^D$, i.e. a bijection φ of $P \cup L$ with $\varphi(P) = L$, $\varphi(L) = P$, preserving

incidence. $H(q)$ has a duality if and only if $q = 3^h$ ([Ti59]; [VM98, Ch. 3]). For $q = 9$ we use an explicit duality found by computer (research/d10D7/dual/dualfind.c: it extends a polarity of the subhexagon $H(3)$ over $\mathbb{F}_3 \subset \mathbb{F}_9$ by propagation along geodesics and then checks that all $N(q+1)$ incidences are preserved).

1.2 The incidence graph and distances

$I = I(q)$ denotes the incidence graph of H: vertex set $P \cup L$ ($2N$ vertices), $x \sim y$ when x and y are incident. $d(x, y)$ is the distance in I, and $\Gamma_i(x) = \{y : d(x, y) = i\}$. Points are at even distance from points, lines at odd distance from points. Two points are *collinear* when $d = 2$, two lines *concurrent* when $d = 2$, and two elements are *opposite* when $d = 6$.

$\Gamma_2(x)$ for a point x is the set of the $q(q+1)$ points collinear with x; for a line ℓ it is the set of the $q(q+1)$ lines concurrent with ℓ. The sizes are $|\Gamma_i(x)| = (q+1)q^{i-1}$ for $1 \le i \le 5$ and $|\Gamma_6(x)| = q^5$.

We use the following standard facts.

(F1) I is connected, bipartite, $(q+1)$-regular, of diameter 6 and girth 12 (this is the definition of a generalized hexagon of order (q, q)).

(F2) Geodesics. If $d(x, y) \le 5$ there is a unique geodesic from x to y (two would close a cycle of length ≤ 10). If $d(x, y) = 6$, then every $z \in \Gamma_1(x)$ satisfies $d(z, y) = 5$ (it is odd, and at least $6 - 1$), so for every neighbour z of x there is exactly one geodesic from x to y through z.

(F3) Covering lemma. Let x, y be elements of H. There is $z \in \Gamma_2(x)$ with $d(z, y) \le 3$ if x and y have different types, and $d(z, y) \le 4$ if they have the same type. *Proof.* If $2 \le d(x, y) \le 6$, take for z the vertex at distance 2 from x on a geodesic from x to y: $d(z, y) = d(x, y) - 2$. If $d(x, y) = 1$, take $z \in \Gamma_1(y) \setminus \{x\}$ (then $z \in \Gamma_2(x)$ and $d(z, y) = 1$). If $x = y$ any $z \in \Gamma_2(x)$ has $d(z, y) = 2$. The largest value is $6 - 2 = 4$ for the same type and $5 - 2 = 3$ for different types. □

(F4) Orthogonality. For points x, y of $H(q)$: $d(x, y) \le 4$ if and only if $b(x, y) = 0$; that is, the points not opposite x are the points of $Q(6, q)$ in the tangent hyperplane $x^\perp$ [VM98, §2.4]. *Proof.* If $d(x, y) \le 4$, there is a point z with x and y on lines of H through z (take $z = x$ if $d(x, y) \le 2$); these lines lie in the totally singular plane π_z of §1.1, so $b(x, y) = 0$. Conversely, both sets have $1+q+q^2+q^3+q^4$ points ($x^\perp \cap Q(6, q)$ is a cone with vertex x over a $Q(4, q)$, and $1+q(q+1)+q^3(q+1)$ points are at distance ≤ 4 from x), so the inclusion is an equality. We re-checked (F4) by computer for all pairs of points for $q = 2, 3, 4$ and for a sample of points for $q = 5$ (Section 10.1); the lab's large computations (research/d13D7/REPORT.md §3) also rely on it.

1.3 Sets of points and lines

Definition 1.1. A set of points of H is

- *non-collinear* if no two of its points are collinear (a partial distance-2 ovoid);
- a *partial ovoid* if its points are pairwise opposite; an *ovoid* if moreover it has $q^3 + 1$ points.

Dually, a set of lines is *non-concurrent*, a *partial spread* (pairwise opposite lines), or a *spread* ($q^3 + 1$ pairwise opposite lines).

By (F4), partial ovoids of $H(q)$ are exactly the sets of pairwise non-orthogonal points of $Q(6, q)$, i.e. partial ovoids of the polar space $Q(6, q)$.

Lemma 1.2 (standard examples).

(a) If Σ is a 3-space of $\mathrm{PG}(6,q)$ with $\Sigma \cap Q(6,q)$ an elliptic quadric $Q^-(3,q)$, its q^2+1 points form a partial ovoid of $H(q)$, for every q.
(b) (Hermitian spread [Th80].) If Π is a hyperplane with $\Pi \cap Q(6,q)$ an elliptic quadric $Q^-(5,q)$, the lines of $H(q)$ contained in Π form a spread, for every q.
(c) If S is a spread, every line not in S meets exactly one line of S.
(d) If $q = 3^h$ and φ is a duality, $\varphi(S)$ is an ovoid for every spread S.

Proof. (a) If $x \neq y$ are points of $Q^-(3,q)$ with $b(x,y) = 0$, then $Q(\lambda x + \mu y) = \lambda\mu\, b(x,y) = 0$, so the line xy lies in $Q^-(3,q)$, which contains no line. So $b(x,y) \neq 0$ and x, y are opposite by (F4).

(b) is in [Th80]; we also checked it by computer for small q (Section 10.1), and the record builders re-check, for the sets they use, the hypotheses of the theorems they apply (Section 10.2).
(c) A line meeting two lines of S would put them at distance ≤ 4. So the lines of S together with the $q(q+1)$ lines meeting each of them are $(q^3+1)(1+q(q+1)) = (q^3+1)(q^2+q+1) = N$ distinct lines, i.e. all lines. (d) φ preserves distances. $\square$

Ovoids of $Q(6,q)$ (equivalently of $H(q)$) exist for $q = 3^h$; they do not exist for q even [Th81] or for q prime ≥ 5 [BGS06]. The largest partial ovoids are not known in general; the lab's record graphs use partial ovoids found by computer (Section 8).

The following class of sets is the one that matters for Theorem C. It contains the non-collinear sets and the point sets of partial spreads.

Definition 1.3 (segment sets). A set F of points is a *segment set* if for every $p \in F$ the points of $F \setminus \{p\}$ that are collinear with p all lie on one line through p. Equivalently: there are no $p, p', p'' \in F$ with $d(p,p') = d(p',p'') = 2$ and $d(p,p'') = 4$ (a path of two collinearities that turns at p'). Dually, a set G of lines is a *dual segment set* if for every $\ell \in G$ the lines of $G \setminus \{\ell\}$ that meet ℓ all pass through one point of ℓ.

So a segment set is a disjoint union of "segments" (sets of ≥ 2 points on a line) and isolated points, such that no two points from different pieces are collinear. Examples:

- every non-collinear set;
- every set in which each point is collinear with at most one other point of the set (the "relaxed sets" (R1) of the lab notes);
- every subset of the point set of a partial spread (points on distinct lines of a partial spread are not collinear, since the lines are opposite);
- dually: the set of all lines through the points of a partial ovoid (lines through distinct points of a partial ovoid never meet).

Proposition 1.4 (size of segment sets). A segment set F has $|F| \leq (q+1)(q^3+1)$, with equality if and only if F is the point set of a spread. Dually, a dual segment set has at most $(q+1)(q^3+1)$ lines, with equality if and only if it is the set of lines through the points of an ovoid (so equality is possible only when $H(q)$ has an ovoid, e.g. $q = 3^h$, and impossible for q even and for q prime ≥ 5).

Proof. Let F_1 be the points of F collinear with no other point of F, and $C_1, \ldots, C_m$ the segments (maximal sets of ≥ 2 points of F on a line ℓ_i; $|C_i| \leq q+1$). A line meeting F in exactly one point is called a *private* line of that point. A point of F_1 has $q+1$ private lines and a point of a segment C_i has q (all its lines except ℓ_i: by the segment property no other line through it contains a second point of F). Private lines of different points are different, and different from the ℓ_i. So

$$(q+1)|F_1| + q\sum_i |C_i| + m \;\leq\; N.$$

Since $m \geq \sum_i |C_i|/(q+1)$, the left side is at least $(q+\frac{1}{q+1})|F|$, which gives $|F| \leq N(q+1)/(q^2+q+1) = (q+1)(q^3+1)$. Equality forces $F_1 = \emptyset$, $|C_i| = q+1$ (whole lines) and $m = q^3+1$. The lines ℓ_i are then pairwise opposite: if ℓ_i and ℓ_k met in a point z, then $z \in F$ would have other points of F on two lines; if $d(\ell_i, \ell_k) = 4$, a point of ℓ_i would be collinear (along the line joining them) with a point of ℓ_k, and would also have other points of F on ℓ_i. So they form a spread. Conversely the point set of a spread is a segment set of this size. The dual statement is the same proof in H^D. $\square$

For comparison, a non-collinear set has at most $N/(q+1) = q^4+q^2+1$ points (each line contains at most one of its points; equality means a distance-2 ovoid, a point set meeting every line exactly once), and $(q+1)(q^3+1) = q^4+q^3+q+1$ is larger by q^3-q^2+q.

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2. Level graphs

Fix $r \geq 1$; there are $2r+1$ levels, indexed by $j \in \mathbb{Z}_{2r+1}$. Each level gets a **type** $\tau_j \in \{H, D\}$. For a type τ let

$$A(H) = P, \quad B(H) = L, \qquad A(D) = L, \quad B(D) = P$$

(the "points" and "lines" of H or of H^D; the letters H, D for the types refer to the hexagon and its dual, not to the diameter). A **transition** is a pair $t = (j, k)$ with $j \in \mathbb{Z}_{2r+1}$, $1 \leq k \leq r$; its source level is j and its target level is $j+k$. It is **friendly** if $\tau_j \neq \tau_{j+k}$; then $A(\tau_j) = B(\tau_{j+k})$.

Definition 2.1 (level graph). Given types τ and, for every transition $t = (j,k)$, a bijection $M_t : A(\tau_j) \to B(\tau_{j+k})$, the graph $X = X(\tau, M)$ has vertex set $(P \cup L) \times \mathbb{Z}_{2r+1}$ and edges

- $(x, j) \sim (y, j)$ whenever x and y are incident ("level edges");
- $(a, j) \sim (M_{(j,k)}(a), j+k)$ for every j, every $1 \leq k \leq r$ and every $a \in A(\tau_j)$ ("matching edges").

Colour the vertices (a, j), $a \in A(\tau_j)$, with colour $\mathcal{A}$ and the vertices (b, j), $b \in B(\tau_j)$, with colour $\mathcal{B}$. Level edges and matching edges join different colours, so X is bipartite. The neighbourhoods are:

$$N_X(a,j) = \Gamma_1(a) \times \{j\} \ \cup \ \{(M_{(j,k)}(a), j+k) : 1 \leq k \leq r\} \qquad (a \in A(\tau_j)),$$

$$N_X(b,j) = \Gamma_1(b) \times \{j\} \ \cup \ \{(M^{-1}_{(j-k,k)}(b), j-k) : 1 \leq k \leq r\} \qquad (b \in B(\tau_j)).$$

So X is $(q+1+r)$-regular of order $(2r+1) \cdot 2N$.

When all levels have type H and all M_t equal one bijection $\sigma : P \to L$, this is the graph whose order is the table value (for $r = 1$: three copies of I joined by $(p, i) \sim (\sigma(p), i+1)$).

Theorem A (level graph). For all types and all bijections M_t, $X(\tau, M)$ has diameter at most 7. More precisely: two vertices of one level are joined by a path of length ≤ 6 inside that level, and for every transition $t = (j,k)$ every vertex of level j is joined to every vertex of level $j+k$ by a path of length ≤ 7 that uses only level edges of levels j and $j+k$ and matching edges of t.

Proof. Inside a level this is (F1). Every two distinct levels differ by $\pm k$ for some $1 \leq k \leq r$ (as there are $2r+1$ levels), so it is enough to join (u, j) to (v, j'), $j' = j+k$, with $M = M_{(j,k)}$:

1. $u \in A(\tau_j)$: $(u, j) \to (M(u), j') \rightsquigarrow (v, j')$, length $\leq 1+6$.
2. $u \in B(\tau_j)$, $v \in B(\tau_{j'})$: with $a = M^{-1}(v)$, $(u, j) \rightsquigarrow (a, j) \to (v, j')$, length $\leq 5+1$ (in I, elements of different types are at distance ≤ 5).
3. $u \in B(\tau_j)$, $v \in A(\tau_{j'})$: pick $b \in \Gamma_1(v)$ and $a = M^{-1}(b)$: $(u, j) \rightsquigarrow (a, j) \to (b, j') \to (v, j')$, length $\leq 5+1+1$. $\square$

3. Surgery and hubs

3.1 The construction

Definition 3.1 (surgery and hubs). Start from a level graph $X(\tau, M)$ and choose for every transition $t = (j, k)$ a set $R_t \subseteq A(\tau_j)$ of **freed elements** (possibly empty).

1. *Deletion.* Delete the matching edges $(a, j) \sim (M_t(a), j + k)$, $a \in R_t$. Call the result X_R. Each deleted edge leaves a **free $\mathcal{A}$-slot** (a, t) at the vertex (a, j) and a **free $\mathcal{B}$-slot** $(M_t(a), t)$ at the vertex $(M_t(a), j + k)$. (An element freed on several transitions out of, or into, a level carries several slots there.) Level j has

$$\alpha_j = \sum_{k=1}^{r} |R_{(j,k)}| \text{ free } \mathcal{A}\text{-slots}, \qquad \beta_j = \sum_{k=1}^{r} |R_{(j-k,k)}| \text{ free } \mathcal{B}\text{-slots}.$$

2. *Hubs.* Add $K_{\mathcal{B}} \leq \min_j \alpha_j$ new vertices of colour $\mathcal{B}$ ($\mathcal{B}$**-hubs**) and $K_{\mathcal{A}} \leq \min_j \beta_j$ new vertices of colour $\mathcal{A}$ ($\mathcal{A}$**-hubs**). Each $\mathcal{B}$-hub h takes exactly one free $\mathcal{A}$-slot in every level j and is joined to its vertex, which we write $(\omega_j(h), j)$; each $\mathcal{A}$-hub g likewise takes one free $\mathcal{B}$-slot in every level, at the vertex $(\omega'_j(g), j)$. Every slot is taken at most once. (The assignment is otherwise arbitrary; a hub may take slots of different freed elements in different levels, and a vertex with several free slots may receive several hubs.)
3. *Hub graph.* Add a bipartite graph Y between the $\mathcal{B}$-hubs and the $\mathcal{A}$-hubs.

Call the result X_R^+. In the lab notes, $\mathcal{B}$-hubs are "line-side hubs" (in an H level they are joined to points) and $\mathcal{A}$-hubs "point-side hubs".

Exact neighbourhoods. For a level vertex, the neighbourhood is that of Definition 2.1 minus the deleted matching edges plus one hub for each taken free slot. For hubs:

$$N_{X_R^+}(h) = \{(\omega_j(h), j) : j \in \mathbb{Z}_{2r+1}\} \cup N_Y(h), \qquad N_{X_R^+}(g) = \{(\omega'_j(g), j) : j \in \mathbb{Z}_{2r+1}\} \cup N_Y(g).$$

Degree and order. Level vertices have degree $\leq q + 1 + r$ (equality when all their free slots are used), hubs have degree $2r + 1 + \deg_Y$. So X_R^+ is bipartite with maximum degree $\leq q + 1 + r$ as soon as Y has maximum degree $\leq q - r$, and its order is

$$|X_R^+| = (2r + 1) \cdot 2N + K_{\mathcal{A}} + K_{\mathcal{B}}.$$

3.2 Assembly

Definition 3.2. A transition $t = (j, k)$ is **repaired** (by R_t) if in X_R every vertex of level j is joined to every vertex of level $j + k$ by a path of length ≤ 7 that uses only level edges of levels $j, j + k$ and the remaining matching edges of t.

Whether t is repaired depends only on M_t and R_t (not on the other transitions, and not on the hubs).

Lemma 3.3 (hubs). Let X_R^+ be as in Definition 3.1. Then

- every hub is within distance 7 of every level vertex;
- every $\mathcal{B}$-hub is within distance 7 of every $\mathcal{A}$-hub.

Proof. A $\mathcal{B}$-hub h and a level vertex (v, j): $h \to (\omega_j(h), j) \rightsquigarrow (v, j)$ inside level j, length $\leq 1+6$. The level copies are never touched by the surgery. A $\mathcal{B}$-hub h and an $\mathcal{A}$-hub g: in level 0, $\omega_0(h) \in A(\tau_0)$ and $\omega'_0(g) \in B(\tau_0)$ have different types, so $h \to (\omega_0(h), 0) \rightsquigarrow (\omega'_0(g), 0) \to g$ has length $\leq 1 + 5 + 1$. □

Theorem 3.4 (assembly). Suppose

- (T) every transition t is repaired by R_t, and
- (HG) any two $\mathcal{B}$-hubs, and any two $\mathcal{A}$-hubs, are within distance 6 in X_R^+.

Then X_R^+ has diameter ≤ 7. If moreover Y has maximum degree $\leq q - r$, then

$$n_b(q + 1 + r,\ 7) \ \geq \ (2r + 1) \cdot 2N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}.$$

Proof. Pairs of level vertices: same level by (F1), different levels by (T) (as in the proof of Theorem A, every pair of levels is the pair of some transition). Pairs with a hub: Lemma 3.3 and (HG). □

(HG) holds for instance when Y alone has all same-colour distances ≤ 6; it can also use paths through the levels (two $\mathcal{B}$-hubs whose slots in some level are at distance ≤ 4 are within distance 6). Section 7 discusses hub graphs.

So everything reduces to geometric conditions under which one transition is repaired. Sections 4 and 5 give two.

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4. Theorem C: transitions that extend to an automorphism

A transition $t = (j, k)$ is **automorphic** if M_t is the restriction to $A(\tau_j)$ of an automorphism ψ of the graph I with $\psi(A(\tau_j)) = B(\tau_{j+k})$. The two cases used here:

- a *friendly* transition with $M_t = \mathrm{id}$ ($\psi = \mathrm{id}$; $A(\tau_j) = B(\tau_{j+k})$);
- an $H \to H$ transition with $M_t = \varphi|_P$ for a duality φ ($q = 3^h$); a $D \to D$ transition with $M_t = \varphi|_L$.

(A friendly transition whose matching is a collineation of H, restricted to points or to lines, is also automorphic.)

Lemma 4.1. Let F be a segment set of points. For all elements u, v of H there is a point $p \notin F$ with

$$d(u, p) + d(p, v) \ \leq \ 6.$$

Proof. If some point p on a geodesic from u to v is not in F, then $d(u, p) + d(p, v) = d(u, v) \leq 6$. Assume that all points on all geodesics from u to v lie in F.

A geodesic with three points p, p', p'' in a row ($p - m - p' - m' - p''$ with lines $m \neq m'$) cannot have all three in F (that is a turning path, excluded by Definition 1.3). A geodesic between two points at distance 4 or 6, a point and a line at distance 5, or two lines at distance 6 contains three points in a row. So only the following cases remain; in each we exhibit $p \notin F$ with $d(u, p) + d(p, v) \leq 6$. We use repeatedly: *if $p_0 \in F$ has a point of F on the line s through it, then no point $p \neq p_0$ collinear with p_0 off s is in F* (Definition 1.3).

(a) $u = v$ a point of F. Let s be a line through u containing every point of F collinear with u (any line if there is none). Take a line $y \neq s$ through u and $p \in y \setminus \{u\}$: $p \notin F$ and $d(u,p) + d(p,u) = 4$.

(b) u, v collinear points of F on the line s. Take $y \neq s$ through u and $p \in y \setminus \{u\}$: $p \notin F$, $d(u,p) = 2$, $d(p,v) \leq 4$.

(c) $u \in F$ a point, v a line through u. If v has a point $p \neq u$ outside F: $2 + 1 = 3$. Otherwise all points of v are in F; take $y \neq v$ through u and $p \in y \setminus \{u\}$: $p \notin F$, $d(u,p) = 2$, $d(p,v) \leq 3$.

(d) u a point, v a line, $d(u,v) = 3$, geodesic $u - s - p_2 - v$ with $u, p_2 \in F$. A point $p \in v \setminus \{p_2\}$ has $d(u,p) = 4$, $d(p,v) = 1$. If all of them were in F, p_2 would have points of F on the two lines s and v.

(e) u, v lines meeting in $p_1 \in F$. Not both u and v contain a second point of F (else p_1 turns). Say u does not: take $p \in u \setminus \{p_1\}$, so $p \notin F$, $d(u,p) = 1$, $d(p,v) = 3$.

(f) u, v lines at distance 4, geodesic $u - p_1 - y - p_3 - v$ with $p_1, p_3 \in F$. Then no other point p of u is in F (else p_1 turns: p on u, p_3 on y), and $d(u,p) = 1$, $d(p,v) \leq d(p,p_1) + d(p_1,v) = 2 + 3$.

(g) $u = v$ a line. If some point of u is not in F: $1 + 1$. Otherwise take $p_1 \in u$, a line $y \neq u$ through p_1 and $p \in y \setminus \{p_1\}$: $p \notin F$ and $d(u,p) = 3$, so $3 + 3 = 6$.

The cases with u a line and v a point are (c), (d) with u and v exchanged. □

Theorem C. Let $t = (j,k)$ be an automorphic transition. If $A(\tau_j) = P$ and R_t is a segment set of points, or $A(\tau_j) = L$ and R_t is a dual segment set of lines, then t is repaired.

Proof. Let ψ be the automorphism of I extending $M = M_t$, $j' = j + k$, and take (u,j), (v,j'). For every $a \in A(\tau_j) \setminus R_t$ the path

$$(u,j) \rightsquigarrow (a,j) \to (\psi(a),j') \rightsquigarrow (v,j')$$

has length $d(u,a) + 1 + d(\psi(a),v) = d(u,a) + 1 + d(a,\psi^{-1}(v))$, since ψ is an isometry of I. If $A(\tau_j) = P$, Lemma 4.1 applied to u, $\psi^{-1}(v)$ and $F = R_t$ gives a with $d(u,a) + d(a,\psi^{-1}(v)) \leq 6$, so the path has length ≤ 7. If $A(\tau_j) = L$, apply Lemma 4.1 in H^D (same graph I, points and lines exchanged; a dual segment set of H is a segment set of H^D). □

Remark 4.2 (the converse). An automorphic transition is repaired (Definition 3.2) *if and only if* its freed set is a segment set (dual segment set for lines). The "if" is Theorem C. For "only if", suppose that $p, p', p'' \in R_t$ turn at p' ($p - m - p' - m' - p''$, $m \neq m'$), take a line $u \neq m'$ through p''. Then $d(p,u) = 5$ with the unique geodesic $p - m - p' - m' - p'' - u$. A path from (u,j) to $(\psi(p),j')$ inside the two levels uses an odd number $2h + 1$ of matching edges of t, and (transporting the segments in level j' back by ψ^{-1} and using the triangle inequality) has length at least $2h + 1 + d(u,p) = 2h + 6$. So a path of length ≤ 7 uses one matching edge, at a point a with $d(u,a) + d(a,p) = 5$, i.e. $a \in \{p, p', p''\}$: all three edges are deleted, and the distance in the two levels is > 7. (In the whole graph, routes through a third level or through hubs might still exist; in the lab's five-level graph over $H(9)$ with the 7,300 spread points freed and no hubs, freeing one more point left 82,720 ordered pairs at distance > 7, research/squads/cells/d12D7.md §2.)

Corollary 4.3. The following transitions are repaired:

1. a friendly transition with $M_t = \mathrm{id}$ and R_t a non-collinear set of points (if $\tau_j = H$) or a non-concurrent set of lines (if $\tau_j = D$) — the "one-transition theorem at friendly transitions" of the lab notes;

2. the same with R_t a set in which every element has at most one partner (collinear, resp. concurrent element of R_t) — the relaxed sets (R1) of research/d15D7b/RELAXED.md, **without** its hub condition (R2);
3. an $H \to H$ transition with M_t a duality and R_t any subset of the points of a partial spread (research/squads/cells/d12D7.md §2; e.g. all $(q+1)(q^3+1)$ points of the Hermitian spread, $q = 3^h$).

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5. Theorem B: local dualities

For transitions that are not automorphic (an $H \to H$ transition when $q \neq 3^h$, or any $H \to H$ transition whose matching is not a duality) we only ask M_t to behave like a duality near the freed elements.

Theorem B (one-transition surgery). Let $t = (j, k)$ be any transition, $M = M_t$, and $R = R_t \subseteq A(\tau_j)$. Suppose

- (H1) no two elements of R are at distance 2 (non-collinear, resp. non-concurrent);
- (H2) no two elements of $M(R)$ are at distance 2;
- (H3) $M(\Gamma_2(x)) = \Gamma_2(M(x))$ for every $x \in R$ (M *is a local duality at* x when M maps points to lines: it maps the points collinear with x onto the lines concurrent with $M(x)$).

Then t is repaired.

Proof. Write $j' = j + k$; "A" and "B" refer to $A(\tau_j), B(\tau_j)$ in level j and to $A(\tau_{j'}), B(\tau_{j'})$ in level j'. Elements of different types are at distance ≤ 5 in I. Take (u, j) and (v, j').

1. $u \in$ A, $u \notin R$: $(u, j) \to (M(u), j') \rightsquigarrow (v, j')$: $\leq 1 + 6$.
2. $u \in$ B, $v \in$ A: the $q + 1 \geq 2$ neighbours of v are pairwise at distance 2, so by (H2) at most one is in $M(R)$; take $b \in \Gamma_1(v) \setminus M(R)$ and $a = M^{-1}(b) \notin R$: $(u, j) \rightsquigarrow (a, j) \to (b, j') \to (v, j')$: $\leq 5 + 1 + 1$.
3. $u \in$ B, $v \in$ B $\setminus M(R)$: $(u, j) \rightsquigarrow (M^{-1}(v), j) \to (v, j')$: $\leq 5 + 1$.
4. $u \in$ B, $v = M(x)$, $x \in R$: by (F3) there is $z \in \Gamma_2(x)$ with $d(z, u) \leq 3$ (x and u have different types). By (H1) $z \notin R$; by (H3) $M(z) \in \Gamma_2(v)$: $(v, j') \rightsquigarrow (M(z), j') \to (z, j) \rightsquigarrow (u, j)$: $\leq 2 + 1 + 3$.
5. $u = x \in R$, $v \in$ A: by (F3) there is $w \in \Gamma_2(M(x))$ with $d(w, v) \leq 3$; by (H3) $w = M(z)$ with $z \in \Gamma_2(x)$, and $z \notin R$ by (H1): $(x, j) \rightsquigarrow (z, j) \to (w, j') \rightsquigarrow (v, j')$: $\leq 2 + 1 + 3$.
6. $u = x \in R$, $v \in$ B: the same path, now with $d(w, v) \leq 4$ by (F3) ($M(x)$ and v have the same type): $\leq 2 + 1 + 4$.

These cases cover all pairs ($u \in$ A $\setminus R$: 1; $u \in R$: 5, 6; $u \in$ B: 2, 3, 4). □

For a friendly transition with $M = \mathrm{id}$, (H3) holds at every element and (H2) is (H1), so Theorem B gives Corollary 4.3.1 again; Theorem C is stronger there. Theorem B is needed for non-automorphic transitions. We now describe its hypotheses for an $H \to H$ transition ($M : P \to L$); the $D \to D$ case is the same with M^{-1}.

For two points x, y at distance 4 let $\mathrm{mid}(x, y)$ be the unique point collinear with both; for two lines at distance 4 let $\mathrm{mid}(\ell, m)$ be the unique line concurrent with both.

Proposition 5.1 (what (H1)–(H3) force). Under (H1)–(H3), M restricted to R preserves distances, and $M(\mathrm{mid}(x, y)) = \mathrm{mid}(M(x), M(y))$ for all $x, y \in R$ at distance 4.

Proof. Let $x \neq y \in R$; $d(x,y) \in \{4,6\}$ by (H1). If $d(x,y) = 4$, $m = \mathrm{mid}(x,y) \in \Gamma_2(x) \cap \Gamma_2(y)$, so by (H3) $M(m) \in \Gamma_2(M(x)) \cap \Gamma_2(M(y))$; hence $d(M(x), M(y)) \leq 4$, $= 4$ by (H2), and $M(m)$ is the mid-line. If $d(x,y) = 6$, $\Gamma_2(x) \cap \Gamma_2(y) = \emptyset$, so $\Gamma_2(M(x)) \cap \Gamma_2(M(y)) = \emptyset$ by (H3), and $d(M(x), M(y)) \notin \{2,4\}$, i.e. $= 6$. □

So $M(R)$ must *mirror* the distance-4 structure of R. The converse, in the form used for the records:

Proposition 5.2 (mirrored sets). Let R be a non-collinear set of points, R^* a non-concurrent set of lines and $f : R \to R^*$ a bijection such that

- (S1) every point is collinear with at most two points of R, and every line is concurrent with at most two lines of R^*;
- (S2) for $x \neq y \in R$: $d(x,y) = 4$ if and only if $d(f(x), f(y)) = 4$.

Then there is a bijection $M : P \to L$ with $M|_R = f$ satisfying (H1)–(H3) for R. One such M is: $M(x) = f(x)$ for $x \in R$; $M(\mathrm{mid}(x,y)) = \mathrm{mid}(f(x), f(y))$ for $x, y \in R$ at distance 4; for each $x \in R$, any bijection from the remaining points of $\Gamma_2(x)$ onto the remaining lines of $\Gamma_2(f(x))$; any bijection on the rest.

Proof. Let $U = \bigcup_{x \in R} \Gamma_2(x)$ and $U^* = \bigcup_{x \in R} \Gamma_2(f(x))$. By (H1) for R and R^*, $R \cap U = \emptyset$ and $R^* \cap U^* = \emptyset$. A point of U lies in $\Gamma_2(x)$ for one or two $x \in R$ (S1); if two, x and y are at distance 4 (not collinear, and both collinear with it) and the point is $\mathrm{mid}(x,y)$. So the points of U in two sets $\Gamma_2(\cdot)$ correspond bijectively to the pairs $\{x,y\} \subseteq R$ at distance 4, and likewise for U^*; by (S2) $\{x,y\} \mapsto \{f(x), f(y)\}$ is a bijection between these pairs. Hence the rule $\mathrm{mid}(x,y) \mapsto \mathrm{mid}(f(x), f(y))$ is a bijection from the shared points of U to the shared lines of U^*, and it maps the shared points of $\Gamma_2(x)$ onto the shared lines of $\Gamma_2(f(x))$. The remaining ("private") points of $\Gamma_2(x)$ and private lines of $\Gamma_2(f(x))$ are then equally many (both $q(q+1)$ minus the number of $y \in R$ at distance 4 from x), and private sets of different x are disjoint. So the pieces define a bijection $R \cup U \to R^* \cup U^*$ with $M(\Gamma_2(x)) = \Gamma_2(f(x))$ for every $x \in R$. As $|P \setminus (R \cup U)| = |L \setminus (R^* \cup U^*)|$, it extends to a bijection $P \to L$. (H1), (H2) are the hypotheses on R and R^*. □

Corollary 5.3.

(a) (Pairwise opposite sets; research/d13D7/REPORT.md.) If R is a partial ovoid and R^* a partial spread with $|R| = |R^*|$, every bijection f satisfies (S1), (S2) (there are no pairs at distance 4).

(b) (Mirrored sets; research/cell147/mirror/.) Let R_0 and X be disjoint partial ovoids, $f_0 : R_0 \to S_0$ a bijection onto a partial spread S_0, and $x \mapsto L_x$ an injective map from X onto a partial spread disjoint from S_0. Put $R = R_0 \cup X$, $R^* = S_0 \cup \{L_x : x \in X\}$, $f = f_0$ on R_0 and $f(x) = L_x$ on X, and assume R non-collinear and R^* non-concurrent. Then (S1) holds automatically: three points of R collinear with one point would be pairwise at distance 4, but two of them lie in R_0 or two in X, which are pairwise opposite; likewise for lines. And (S2) says: for $r \in R_0$ and $x \in X$, $d(r,x) = 4$ if and only if $d(f_0(r), L_x) = 4$.

In (b) the matching condition is a bipartite assignment problem: each $r \in R_0$ has a "type", the set of $x \in X$ at distance 4 from it, and must be sent to a spread line of the same type with respect to the lines L_x. The record sets were found by annealing over X, L_x and f (research/cell147/mirror/mirror_sa*.py [b14]), and every one was re-checked by the builder (non-collinearity, non-concurrency, and (H3) at every $x \in R$, directly).

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6. Typed level layouts and slot counts

By Theorem 3.4 the order is $(2r+1)\cdot 2N + K_{\mathcal{A}} + K_{\mathcal{B}}$ with $K_{\mathcal{B}} \le \min_j \alpha_j$ and $K_{\mathcal{A}} \le \min_j \beta_j$, where α_j (β_j) counts the freed elements of the transitions leaving (entering) level j. Which transitions are friendly depends on the types.

All levels of type H, one bijection σ, one freed set R on every transition (the three-level records (7,7), (9,7), (10,7), (13,7) with σ a local duality on R, and the duality records (11,7) with three levels and (12,7) with five). Every level has $\alpha_j = \beta_j = r|R|$, so $K_{\mathcal{A}} = K_{\mathcal{B}} = r|R|$ and

$$|X_R^+| = (2r+1)\cdot 2N + 2r|R|.$$

Hubs: for $r = 1$ the $\mathcal{B}$-hub h_x of $x \in R$ is joined to $(x,0),(x,1),(x,2)$ and the $\mathcal{A}$-hub g_x to $(\sigma(x),0),(\sigma(x),1),(\sigma(x),2)$, plus $q-1$ hub-graph edges each.

Types $HDHDH$ ($r = 2$). Transitions $j \to j+1$ and $j \to j+2$. Friendly: $0 \to 1$, $2 \to 3$, $4 \to 1$ ($H \to D$, points freed) and $1 \to 2$, $3 \to 4$, $3 \to 0$ ($D \to H$, lines freed). Not friendly: $0 \to 2$, $2 \to 4$, $4 \to 0$ ($H \to H$) and $1 \to 3$ ($D \to D$). With F_t the freed sets:

level	type	free $\mathcal{A}$-slots α_j	free $\mathcal{B}$-slots β_j
0	H	$F_{0\to1} + R_{0\to2}$	$R_{4\to0} + F_{3\to0}$
1	D	$F_{1\to2} + R_{1\to3}$	$F_{0\to1} + F_{4\to1}$
2	H	$F_{2\to3} + R_{2\to4}$	$F_{1\to2} + R_{0\to2}$
3	D	$F_{3\to4} + F_{3\to0}$	$F_{2\to3} + R_{1\to3}$
4	H	$R_{4\to0} + F_{4\to1}$	$F_{3\to4} + R_{2\to4}$

(sizes of the sets). With all point sets of size f_P, all line sets of size f_L, and m elements freed on each non-friendly transition: $K_{\mathcal{B}} \le \min(f_P + m,\ f_L + m,\ 2f_L)$ and $K_{\mathcal{A}} \le \min(f_L + m,\ 2f_P,\ f_P + m)$. Five levels and two matchings form the regular tournament on 5 vertices, and a split of the types into $3 + 2$ levels cuts 6 of the 10 arcs, the maximum. The split $HDHDH$ (the two D levels at distance 2) also gives every level at least one friendly outgoing and one friendly incoming transition; a split with adjacent D levels, such as $DDHHH$, does not. (With three levels, some level always has no friendly outgoing transition, so typing does not help for $r = 1$.)

Seven levels ($r = 3$). The (15,7) graph uses types $HHDHDHD$: 12 of the 21 transitions are friendly and every level has at least one friendly transition in each direction (research/d15D7b/README.md [b15]).

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7. Hub graphs

Theorem 3.4 needs a bipartite hub graph Y of maximum degree $\le q - r$ (for the degree bound) such that same-colour hubs are within distance 6 in X_R^+ (condition (HG); paths through the levels may help). This is a purely combinatorial side condition. Write $\mathrm{HG}(K_1, K_2; \delta)$ for a bipartite graph with colour classes of sizes K_1, K_2, maximum degree $\le \delta$, and all same-colour distances ≤ 6 *in the graph itself*; if Y is such a graph with $\delta \le q - r$, both conditions hold.

Lemma 7.1 (plane with pendant trees). Let s be a prime power and $t, u \ge 0$ integers with $s + 1 + t \le \delta$ and $u + 1 \le \delta$. Then $\mathrm{HG}(K, K; \delta)$ exists for every K with $s^2 + s + 1 \le K \le (s^2 + s + 1)(1 + t + tu)$.

Proof. Start from the incidence graph of $\mathrm{PG}(2,s)$ (the *core*: diameter 3, same-colour distance 2). Attach to some core vertices up to t pendant vertices each ("depth 1", of the opposite colour), and to some depth-1 vertices up to u pendant vertices each ("depth 2", of the colour of the core vertex). Degrees are at most $s+1+t$, $u+1$ and 1. For two vertices x, y of the same colour with core ancestors $c(x), c(y)$, $d(x,y) \le \mathrm{depth}(x) + d(c(x),c(y)) + \mathrm{depth}(y)$. The core ancestors have the same colour when the depths have the same parity (core distance ≤ 2) and different colours otherwise (core distance ≤ 3); the possible totals are $0+2+0$, $0+3+1$, $0+2+2$, $1+2+1$, $1+3+2$, $2+2+2$, all ≤ 6. To get K vertices of each colour, give c depth-1 children to the points and c to the lines, and g depth-2 children below each of these two families, with $c = \lceil (K - s^2 - s - 1)/(1+u) \rceil \le (s^2+s+1)t$ and $g = K - s^2 - s - 1 - c \le uc$. □

Taking $t = \delta - s - 1$ and $u = \delta - 1$, Lemma 7.1 gives hub graphs with up to $\kappa(\delta) := \max_s (s^2+s+1)(1+(\delta-s-1)\delta) \approx \frac{4}{27}\delta^4$ vertices per colour (s over prime powers $\le \delta - 1$; $\kappa(3) = 7$, $\kappa(4) = 35$, $\kappa(8) = 527$, $\kappa(9) = 868$, $\kappa(10) = 1{,}271$, $\kappa(11) = 1{,}938$, $\kappa(14) = 5{,}187$). For $\delta \ge 3$ the intervals for $s = 2$ and for the optimal s overlap, so $\mathrm{HG}(K, K; \delta)$ exists for every $7 \le K \le \kappa(\delta)$. This is enough for the general statements of Section 9 but far below what the records need. The record hub graphs were found by computer and checked:

cell	hubs per colour	hub de-gree	hub graph	(HG) checked by
(7,7), (9,7), (10,7), (13,7)	81, 295, 229, 768	$q-1$	random $(q-1)$-regular, retried until same-colour distances ≤ 6 in Y	research/d13D7/multi.c (BFS in Y), then the full all-pairs check
(11,7)	7,300	8	Haar graph of $\mathbb{Z}_{73} \rtimes \mathbb{Z}_{100}$, diameter 7	build_haar2.py, then full check
(12,7)	14,600	7	Haar graph of $(\mathbb{Z}_{73} \rtimes \mathbb{Z}_8) \times \mathbb{Z}_5^2$, with paths through the levels ("cluster-assisted", research/squads/cells/d12D7.md §2)	full check
(14,7)	9,014	9	9 random perfect matchings	BFS in Y, then full check
(15,7)	10,826 / 10,481	≤ 8	random, repaired by 2-switches; paths through the levels allowed	BFS from all hubs, then full check
(16,7)	14,884	≤ 11	voltage graph over $\mathbb{Z}_{61}$	hubcheck.c, then full check

8. The record graphs as corollaries

In each case below the graph is X_R^+ of Definition 3.1, every transition is repaired by Theorem B or C, and (HG) holds, so Theorem 3.4 gives the bound. "Computer" marks the parts that are found by search and checked by computer rather than proved in general: the specific sets, the bijection f, and the hub graph. The theorems themselves are proved for all q.

Corollary 8.1 (three levels, one local duality: (7,7), (9,7), (10,7), (13,7)). Let $r = 1$, all types H, $M_t = \sigma$ for all three transitions and $R_t = R$ for all three, with (R, σ) as in Proposition 5.2 / Corollary 5.3(b). If the hub graph Y (with $|R|$ hubs of each colour) has maximum degree $\le q - 1$

and satisfies (HG) of Theorem 3.4 (for instance, if it is an HG($|R|, |R|; q-1$)), then

$$n_b(q+2,\,7) \;\geq\; 6N(q) + 2|R|.$$

Records: $q = 5$: $|R| = 81$ ($R_0 = 72$ pairwise opposite points, 9 extra points, spread lines of $q^3+1 = 126$ available), $n = 23{,}598$; $q = 7$: $288+7$, $n = 118{,}238$; $q = 8$: $217+12$, $n = 225{,}152$; $q = 11$: $756 + 12$, $n = 1{,}064{,}472$. Computer: R_0 (for $q = 7, 11$ an orbit of the Weyl group $W(E_7)$ acting on $Q(6, q)$ through the E_7 root lattice mod q, research/cell147/ovoid/ and octo/ [b14]; being pairwise non-orthogonal in $Q(6, q)$ it is a partial ovoid of $H(q)$ by (F4); for $q = 5, 8$ from research/d13D7/q57 and q8 [b14]), the extra points X, the lines L_x, the bijection f onto Hermitian spread lines, the hub graph.

Corollary 8.2 (duality, $q = 3^h$: (11,7), (12,7)). Let $q = 3^h$, φ a duality, all types H, $M_t = \varphi|_P$ for every transition, and $R_t = F$ for every transition, F the point set of a spread (e.g. the Hermitian spread), $|F| = (q+1)(q^3+1)$. By Theorem C every transition is repaired. Every level has $r|F|$ free slots of each colour. If $K \leq r(q+1)(q^3+1)$ hubs of each colour are attached and the hub graph Y has maximum degree $\leq q - r$ and satisfies (HG) (for instance, if it is an HG($K, K; q-r$)), then

$$n_b(q+1+r,\,7) \;\geq\; (2r+1)\cdot 2N(q) + 2K.$$

Records ($q = 9$, $|F| = 7{,}300$): $r = 1$, $K = 7{,}300$: $n = 413{,}180$; $r = 2$, $K = 14{,}600$: $n = 693{,}500$. By Proposition 1.4 and Remark 4.2 no larger freed set leaves every transition repaired, so within this construction, and as long as condition (T) of Theorem 3.4 is required, $K = r(q+1)(q^3+1)$ is the cap. Computer: the duality φ of $H(9)$ (explicit, verified), the hub graphs. Proved: the freed set and its size, for every $q = 3^h$.

Corollary 8.3 (typed levels: (14,7), (15,7), (16,7)). Let the types be $HDHDH$ ($r = 2$) or $HHDHDHD$ ($r = 3$), let every friendly transition carry $M_t = \mathrm{id}$ and a segment set (points) or dual segment set (lines) F_t, and let the non-friendly transitions carry $M_t = \sigma$ ($H \to H$) or σ^{-1} ($D \to D$) with freed sets R (resp. $\sigma(R)$) satisfying Theorem B. If $K_{\mathcal{B}} \leq \min \alpha_j$, $K_{\mathcal{A}} \leq \min \beta_j$ (Section 6) and the hub graph Y has maximum degree $\leq q - r$ and satisfies (HG) (for instance, if it is an HG($K_{\mathcal{B}}, K_{\mathcal{A}}; q-r$)), then

$$n_b(q+1+r,\,7) \;\geq\; (2r+1)\cdot 2N(q) + K_{\mathcal{A}} + K_{\mathcal{B}}.$$

Records:

- (14,7), $q = 11$, $HDHDH$: all friendly point sets equal one non-collinear set of 8,645 points, all friendly line sets one non-concurrent set of 8,246 lines (both unions of orbits of an order-133 collineation), and the 768-point mirrored set of Corollary 8.1 on the four non-friendly transitions. $\alpha = (9413, 9014, 9413, 16492, 9413)$, $\beta = (9014, 17290, 9014, 9413, 9014)$, $K_{\mathcal{A}} = K_{\mathcal{B}} = 9{,}014$: $n = 1{,}789{,}588$.
- (15,7), $q = 11$, $HHDHDHD$: twelve friendly sets with 5,674–9,303 elements each; in the four bottleneck sets some elements have one partner (5,577 pairs in all), so these are segment sets but not non-collinear; the 768-point mirrored set on the nine non-friendly transitions. $K_{\mathcal{B}} = 10{,}826$, $K_{\mathcal{A}} = 10{,}481$: $n = 2{,}501{,}491$. (The graph also satisfies the lab's unnecessary hub condition (R2).)
- (16,7), $q = 13$, $HDHDH$: a non-collinear set of 14,762 points and a non-concurrent set of 14,823 lines (unions of orbits of an order-61 element of $SL(3, 13)$), and on the four non-friendly transitions a pairwise opposite set of 122 points (resp. lines) with σ as in Corollary 5.3(a). $K_{\mathcal{A}} = K_{\mathcal{B}} = 14{,}762 + 122 = 14{,}884$: $n = 4{,}052{,}108$.

Computer: all these sets, σ, and the hub graphs.

8.4 Where the exact graphs are. Each record graph is determined by a solution file (research/frontier/sols/ on master) that lists the freed sets, the matchings (or the data that determine them), the slot $\omega_j(h)$ of every hub in every level, and the hub graph Y; the builder named here turns it into an explicit edge list (which the coordinator then checked, Section 10.2). So the neighbourhood of every vertex, including every hub, can be read off.

cell	solution file (research/frontier/sols/)	builder
(7,7)	d7D7_hex5_mirror81_n23598.sol	research/d13D7/tools/multi_sol.py
(9,7)	d9D7_hex7_mirror295_n118238.sol	same
(10,7)	d10D7_hex8_mirror229_n225152.sol	same
(11,7)	d11D7_hex9_spread730_haarMeta73x100_n413180.sol, .hubs	research/d10D7/dual/xgen.c
(12,7)	d12D7_hex9_spread730_clhaarZ73Z8xZ5sq_n693500.sol, .hubs3	research/squads/d12D7/clhaar/x5gen
(13,7)	d13D7_hex11_mirror768_n1064472.sol	research/d13D7/tools/multi_sol.py
(14,7)	d14D7_hex11_typedmirror768_n1789588.sol	research/cell147/typedmirror/tm_bu
(15,7)	d15D7_hex11_typedrelax2_n2501491.sol	research/d15D7b/tm3_build.py
(16,7)	d16D7_hex13_typed61_K244_n4052108.sol	research/squads/cells/d16D7/typed5s

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9. What is proved for every q

Theorems A, B, C and 3.4 hold for every prime power q (the proofs only use (F1)–(F3), and at least two points on a line and two lines on a point). What is not known in general is how large the sets can be made and whether large hub graphs exist; Lemma 1.2 and Lemma 7.1 give explicit choices. With $\kappa(\delta)$ as in Section 7:

Corollary 9.1 (every $q \geq 4$, three levels). With R the q^2+1 points of an elliptic quadric $Q^-(3,q)$ section (Lemma 1.2(a)) and R^* any q^2+1 lines of the Hermitian spread (Corollary 5.3(a)):

$$n_b(q+2,\,7) \;\geq\; 6N(q) + 2\min\bigl(q^2+1,\;\kappa(q-1)\bigr),$$

which is $6N(q)+2(q^2+1)$ for every $q \geq 5$.

Corollary 9.2 ($q = 3^h$). For every r with $1 \leq r \leq q-3$ (Corollary 8.2 with Lemma 7.1; here $\kappa(q-r) < q^4 < r(q+1)(q^3+1)$, so the hub graph is the only limit):

$$n_b(q+1+r,\,7) \;\geq\; (2r+1)\cdot 2N(q) + 2\kappa(q-r).$$

Corollary 9.3 (every $q \geq 5$, typed levels $HDHDH$). Friendly point sets: the point set of the Hermitian spread ($(q+1)(q^3+1)$ points); friendly line sets: the lines through the points of an elliptic quadric section ($(q+1)(q^2+1)$ lines, a dual segment set); non-friendly: $m = q^2+1$ as in Corollary 9.1. By the table of Section 6 the smallest slot counts are $\min_j \alpha_j = \min_j \beta_j = (q+1)(q^2+1) + q^2 + 1 = (q+2)(q^2+1)$, so

$$n_b(q+3,\,7) \;\geq\; 10N(q) + 2\min\bigl((q+2)(q^2+1),\;\kappa(q-2)\bigr).$$

The minimum is $\kappa(q-2)$ for $q \leq 13$ and $(q+2)(q^2+1)$ for $q \geq 16$.

The record values exceed these because the sets found by computer are of order q^4 (non-collinear sets of 8,645 and 14,762 points at $q = 11, 13$, against $q^4 = 14{,}641,\ 28{,}561$) and the hub graphs found by computer are much larger than κ. For $q = 3^h$ the sets are already optimal for Theorem C (Proposition 1.4) and only the hub graphs are found by computer.

Remark 9.4 (consequences of Theorem C not yet built). Theorem C allows friendly sets that are much larger than the non-collinear and non-concurrent sets used in the (14,7)–(16,7) records: up to $(q+1)(q^3+1)$ (Proposition 1.4), e.g. the 15,984 points of the Hermitian spread of $H(11)$ against 8,645, and the $12 \cdot 756 = 9{,}072$ lines through the 756-point partial ovoid of $H(11)$ against 8,246 (non-concurrent). In the $HDHDH$ layout with these sets and the 768 mirrored elements the slot counts of Section 6 give $K_{\mathcal{A}} = K_{\mathcal{B}} = 9{,}072 + 768 = 9{,}840$, i.e. $n = 1{,}771{,}560 + 19{,}680 = 1{,}791{,}240$ at (14,7), provided a hub graph with 9,840 hubs per colour and degree 9 exists. We have not built or checked these graphs; they are left to the lab's search sessions.

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10. Computer checks

10.1 Small-q tests of every theorem

Two independent sets of programs (details, commands and logs in the two README files):

- research/theorems/d7hex_tests/: builds $H(q)$ from scratch from the coordinates of Section 1.1 ($q = 2, 3, 4, 5$), builds each graph from Definitions 2.1 and 3.1 (asserting that every matching is a bijection), and checks it with research/lifts/coordinator_check/apsp_mt.c (breadth-first search from every vertex; "VALID" = bipartite, maximum degree $\leq \Delta$, diameter ≤ 7);
- research/theorems/d7hex_lemma41/: uses the lab's generator research/d10D7/hexgen2.c and duality finder research/d10D7/dual/dualfind.c; checks Lemma 4.1 directly on all ordered pairs (u, v), and the duality construction.

statement	what was built	q	result
§1.1 coordinates, (F1), (F4), Lemma 1.2(b)	$H(q)$ from Tits' equations; counts, diameter 6, girth 12 (distance profile), (F4) on all pairs ($q \leq 4$) or a sample ($q = 5$); Hermitian spread in $x_0 = cx_4 + ex_3$	2, 3, 4, 5	all hold; spreads of 28, 65, 126 pairwise opposite lines
Theorem A	plain level graphs with random bijections, types HHH, HDH, $HHHHH$, $HDHDH$ (and $HHDHHDD$, $HHHHHHH$ for $q = 3, 4$)	3, 4, 5	all VALID, $(q+1+r)$-regular
Lemma 4.1	all ordered pairs, for spread-point sets, three random maximal segment sets, a random maximal (R1) set; points and lines	3, 4, 5	0 failures; each set plus one turning point: 36–320 failing pairs
Theorem C (friendly)	two levels $H \to D$, $D \to H$ and five levels $HDHDH$, main graph without hubs, with: all points of the Hermitian spread (112, 325, 756 $= (q+1)(q^3+1)$), random maximal segment sets, (R1) sets with no hub condition, pencils of a partial ovoid, dual segment sets; 3 seeds	3, 4, 5	108 of 108 VALID; set plus one turning point: 72–1,800 ordered pairs at distance > 7 (apsp_mt: diameter 8, INVALID)

statement	what was built	q	result
Theorem C (friendly, larger q)	two levels $H \to D$, all $(q+1)(q^3+1)$ points of the Hermitian spread freed (2,752 and 15,984 points), main graph	7, 11	VALID (n = 78,432 and 708,624; all sources)
Theorem C (duality), Cor. 8.2	all levels H, every transition matched by a polarity φ of $H(3)$, the 112 spread points freed on every transition, $r = 1, 2$	3	VALID (n = 2,184 and 3,640); one more point freed: diameter 8
Theorem B, Prop. 5.2, Cor. 5.3	M built exactly as in Proposition 5.2, (H1)–(H3) re-checked on it; R pairwise opposite (an ovoid of 28 points at $q = 3$; 23; 33) and mirrored R with pairs at distance 4 (17 points/19 pairs; 26/21; 38/28); types HHH (same M, R on all transitions) and $HDHDH$ (M on $H \to H$, M^{-1} on $D \to D$)	3, 4, 5	all VALID; breaking (H3) at one point: 50, 76, 126 ordered pairs at distance > 7
Theorem 3.4 (hubs)	full graphs with hubs on random free slots and a random hub graph of degree $\leq q - r$: HHH, HDH (K = slot minimum, 28 and 35), $HDHDH$ (K = 48 and 61, below the slot minimum because the hub degree $q - 2$ is small)	4, 5	all VALID (n = 8,246, 8,246, 13,746; 23,506, 23,506, 39,182)

Every positive test passed and every negative control failed. Small q cannot test the hub graphs at the record sizes (at $q \leq 5$ the hub degree $q - r$ is 1 to 4); for those see Section 10.2.

10.2 The record graphs

Each record graph was rebuilt by the lab's coordinator (on a separate machine) from its solution file with the session's builder, which also re-checks the hypotheses ((H1)–(H3) at every freed element, non-collinearity or the partner condition of the friendly sets, degrees, the hub graph). The edge list was then checked by research/lifts/coordinator_check/apsp_mt.c: simple graph, maximum degree $\leq \Delta$, bipartite, connected, and a bit-parallel breadth-first search from **every** vertex, no symmetry assumed. All nine runs are in research/coordinator_check/check_2026-10-25.txt:

cell	n	apsp_mt output (all sources)
(7,7)	23,598	RESULT VALID d=7 D=7 n=23598
(9,7)	118,238	RESULT VALID d=9 D=7 n=118238
(10,7)	225,152	RESULT VALID d=10 D=7 n=225152
(11,7)	413,180	RESULT VALID d=11 D=7 n=413180
(12,7)	693,500	RESULT VALID d=12 D=7 n=693500
(13,7)	1,064,472	RESULT VALID d=13 D=7 n=1064472
(14,7)	1,789,588	RESULT VALID d=14 D=7 n=1789588
(15,7)	2,501,491	RESULT VALID d=15 D=7 n=2501491
(16,7)	4,052,108	RESULT VALID d=16 D=7 n=4052108

(The finders' own checks were partly reduced by symmetry, e.g. an order-61 automorphism for (16,7); the coordinator's runs above assume none.)

11. Prior art, and what is new

Classical. Generalized hexagons and their incidence graphs [Ti59]; the split Cayley hexagon in $Q(6,q)$, its dualities for $q = 3^h$, and the Hermitian spread ([Ti59], [Th80], [VM98]).

The level graph of Definition 2.1 is a bipartite analogue of the *star product* of Bermond, Delorme and Farhi [BDF82, BDF84] (see [MS13, §3.3.2]): copies of I indexed by the vertices of the complete graph K_{2r+1}, oriented as a regular tournament, and for each arc a perfect matching between one colour class of the tail copy and the other colour class of the head copy. The survey credits the constructions of large bipartite graphs mainly to Delorme's bipartite versions of the star product and of compounding [De85a, De85b], to the partial Cayley graphs of Bond and Delorme, and to the voltage lifts of Loz, Pérez-Rosés and Pineda-Villavicencio, which updated about a third of the table [MS13, §3.4.5]. The D = 7 values with $\Delta \geq 9$, and (7,7), have exactly the form $(2r+1) \cdot 2N(q)$ of Theorem A; the survey's table does not name the source of each cell, and we have not determined it. We do not claim Theorem A as new: it is the routine argument for such level graphs.

New here (as far as we know). Deleting matching edges at freed elements and reattaching the freed slots to hub vertices (Definition 3.1, Theorem 3.4); the repair theorems B and C; the use of dual levels ("typed levels") to make transitions automorphic; segment sets and Proposition 1.4; mirrored sets (Propositions 5.1, 5.2); and the nine values in Section 0.

12. Appendix: notes for readers of the lab notes

What is new relative to the per-cell notes. Writing the proofs out gave one simplification: Theorem C replaces the lab's conditions on freed sets at friendly transitions (non-collinear sets, and the "relaxed sets" with hypotheses (R1), (R2) of research/d15D7b/RELAXED.md) by the single, weaker condition of being a segment set. The lab's (12,7) notes (research/squads/cells/d12D7.md §2) state the same condition for transitions matched by a duality and check it by computer at $q = 9$; Theorem C proves it and transfers it to friendly transitions. The record graphs are unaffected (they satisfy the stronger hypotheses and were checked by computer in full), but larger sets are now allowed (Remark 9.4).

Why the lab's (R2) was not needed. The lab proof of the relaxed theorem (RELAXED.md, cases 4 and 5) only used paths whose matching edge sits at a point collinear with the freed point r. When r has a partner x and the target is a line u with $d(r,u) = 5$ and geodesic $r - y - x - y_2 - w_2 - u$, the only such point is x (freed), and (R2) supplies a hub. The path through w_2 (at distance 4 from r) avoids this: w_2 is collinear with x, so it is not freed (else x would have two partners), and $(r, j') \rightsquigarrow (w_2, j') \to (w_2, j) \to (u, j)$ has length $4 + 1 + 1 = 6$. This is the first step of the proof of Lemma 4.1 (a point and a line at distance 5: three points in a row on the geodesic).

Hubs in Theorem B. The earlier version of Theorem B (research/d13D7/REPORT.md §2, with R pairwise opposite) routed case 6 through the hub of x in two levels. The route of case 6 in the proof of Theorem B uses no hub, which is what lets hubs take their slots from different freed elements in different levels (Definition 3.1).

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